

RUIBO CHEN

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EDUCATION

University of Maryland, College Park, U.S.

Ph.D. in Computer Science. Advisor: Prof. Heng Huang

Aug. 2023 – Present

Peking University, China

B.S. in Intelligence Science and Technology. GPA: 3.813/4.0, Ranking: 4th/81

Aug. 2019 – Jul. 2023

SELECTED PREPRINTS AND PUBLICATIONS

A full list of my publications is available on my [Google Scholar](#) profile.

*denotes equal contribution.

• VISUAL GENERATION AND MULTIMODAL LEARNING

[1] POET: Preference Optimization for Enhanced Text-to-Image Generation

Ruibo Chen*, Jiacheng Pan*, Heng Huang, Zhenheng Yang. *ECCV 2026*. [\[Paper\]](#)

◦ Address the training–inference mismatch in text-to-image generation, where short user prompts deviate from the detailed captions that models are trained on. Use iterative preference optimization to train an LLM to rewrite user prompts to match the T2I model’s input distribution, improving image quality, aesthetics, and text-image alignment.

[2] Scale as a Modality: Multi-Scale Pixel-Space Diffusion Transformers

Ruibo Chen, Kaishen Wang, Tianyi Xiong, Chenxi Liu, Yanshuo Chen, Heng Huang, Jian Du. *Under Review*.

◦ Introduce MS-pDiT, a VAE-free multi-scale pixel-space diffusion transformer that tokenizes the same image at multiple patch sizes and couples per-scale streams via joint attention, making the number of scales a new scaling dimension. Achieve FID 1.43 / 1.44 on ImageNet 256 / 512 and 0.86 on GenEval.

[3] Your Vision-Language Model Itself Is a Strong Filter: Towards High-Quality Instruction Tuning with Data Selection

Ruibo Chen, Yihan Wu, Lichang Chen, Guodong Liu, Qi He, Tianyi Xiong, Chenxi Liu, Junfeng Guo, Heng Huang. *ACL 2024 (Findings)*. [\[Paper\]](#)

◦ Improve instruction tuning by using large vision-language models themselves to select high-quality data. The first to show that, for large-scale multimodal instruction tuning, a small amount of high-quality data is sufficient.

• WATERMARKING AND TRUSTWORTHY GENERATIVE AI

[1] OriginMark: Mitigating Attacks in Distortion-Free Watermarks

Ruibo Chen, Donghang Lu, Heng Huang, Jian Du. *Under Review*.

◦ Reveal that fixed-key distortion-free watermarks leak an exploitable fingerprint, enabling black-box identification and spoofing attacks. Propose OriginMark, which neutralizes these attacks with only eight keys while outperforming attack-resistant baselines in detectability.

[2] TTMark: Pairwise Distortion-Free Watermarking Beyond Single-Token Entropy

Ruibo Chen, Zhengmian Hu, Donghang Lu, Xuehao Cui, Georgios Milis, Yihan Wu, Jian Du, Heng Huang. *Under Review*. [\[Paper\]](#)

◦ Propose a pairwise framework that applies distortion-free watermarks to the joint distribution of adjacent tokens, overcoming the single-token entropy bottleneck. Provably and empirically improve detectability without degrading generation quality.

[3] More Haste, Less Speed: Weaker Single-Layer Watermark Improves Distortion-Free Watermark Ensembles

Ruibo Chen, Yihan Wu, Xuehao Cui, Jingqi Zhang, Heng Huang. *EMNLP 2026*. [\[Paper\]](#)

◦ Show that strong per-layer watermarks reduce token entropy and weaken subsequent layers in watermark ensembles, and propose using weaker single-layer watermarks to preserve entropy, consistently improving detectability.

[4] MC²Mark: Distortion-Free Multi-Bit Watermarking for Long Messages

Xuehao Cui*, **Ruibo Chen***, Yihan Wu, Heng Huang. *NeurIPS 2026*. [\[Paper\]](#)

- Propose a distortion-free multi-bit watermark that encodes long messages via multi-channel colored reweighting, achieving near-perfect accuracy on short messages and greatly outperforming baselines on long messages.

[5] An Ensemble Framework for Unbiased Language Model Watermarking

Yihan Wu*, **Ruibo Chen***, Georgios Milis, Heng Huang. *ICLR 2026*. [\[Paper\]](#)

- Introduce a novel ensemble framework that enhances the detectability and robustness of unbiased large language model watermarking by sequentially composing multiple independent watermark instances.

[6] De-mark: Watermark Removal in Large Language Models

Ruibo Chen*, Yihan Wu*, Junfeng Guo, Heng Huang. *ICML 2025*. [\[Paper\]](#)

- Effectively remove and exploit the popular n-gram-based watermarks of large language models by reverse-engineering the watermarking parameters with a provably unbiased estimator and theoretical guarantees for the distribution gap.

[7] Improved Unbiased Watermark for Large Language Models

Ruibo Chen*, Yihan Wu*, Junfeng Guo, Heng Huang. *ACL 2025*. [\[Paper\]](#)

- Propose MCmark, a multi-channel unbiased watermark that achieves state-of-the-art detectability across various models and datasets, with theoretical guarantees of improvement over prior unbiased watermarks.
- OpenAI's [textGrain](#) watermark, being deployed in ChatGPT and Codex, is an improved variant of this framework.

[8] A Watermark for Order-Agnostic Language Models

Ruibo Chen*, Yihan Wu*, Yanshuo Chen, Chenxi Liu, Junfeng Guo, Heng Huang. *ICLR 2025*. [\[Paper\]](#)

- Propose the first watermarking framework for order-agnostic language models (e.g., diffusion language models), with a novel and more effective pattern-based detection algorithm.

[9] Robust Distortion-Free Watermark for Autoregressive Audio Generation Models

Yihan Wu*, Georgios Milis*, **Ruibo Chen***, Heng Huang. *NeurIPS 2025*. [\[Paper\]](#)

- Introduce a novel distortion-free watermarking method for autoregressive audio generation models that overcomes retokenization mismatch through clustering-based aligned inverse sampling, achieving high detectability and robustness without degrading audio quality.

EXPERIENCE

- *Research Intern* | **TikTok** 2026
Mentor: Jian Du San Jose, CA
 - **Developed the core watermarking algorithms for ByteDance LLM products.** Conducted research on improving the detectability and security of distortion-free LLM watermarks. [\[TTMark, Under Review\]](#) [\[OriginMark, Under Review\]](#)
 - Built and pretrained VAE-free multi-scale pixel-space diffusion transformers for class-conditional and text-to-image generation. [\[MS-pDiT, Under Review\]](#)
- *Research Intern* | **TikTok** 2025
Mentor: Jiacheng Pan, Zhenheng Yang San Jose, CA
 - Mitigated the training-inference mismatch in text-to-image models by training an LLM with RL to align user prompts with the model's input distribution, improving generation quality. [\[POET, ECCV'26\]](#)

SELECTED SCHOLARSHIPS & AWARDS

- **Dean's Fellowship**, University of Maryland, College Park, 2023-2025
- **Outstanding Graduates of Beijing**, 2023
- **Outstanding Graduates of Peking University**, 2023
- **National Scholarship**, China, 2022